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# Privacy-Preserving Federated Learning Framework for DDoS Attack Detection in IoT Networks Using Hybrid Feature Selection

**HARIS MOHAMED C P[1], ANINDITA KUNDU[2]**
[1]School of Computer Science and Engineering, Vellore Institute of Technology, Vellore, Tamil Nadu, India (e-mail: harismohamedcp@gmail.com)
[2]School of Computer Science and Engineering, Vellore Institute of Technology, Vellore, Tamil Nadu, India (e-mail: anindita.kundu@vit.ac.in)

Corresponding author: Anindita Kundu (e-mail: anindita.kundu@vit.ac.in).

**ABSTRACT** As Internet of Things (IoT) deployments scale across sectors, Distributed Denial of Service (DDoS) attacks have grown into one of the most disruptive security threats they face—capable of disabling infrastructure without any physical access. Defending against these attacks through centralized machine learning is problematic: it requires pooling sensitive network traffic from distributed nodes onto a single server, creating privacy liabilities and a structurally fragile single point of failure. We address both issues simultaneously through a federated learning framework that trains Support Vector Machine (SVM) classifiers locally across distributed IoT workers while sharing only aggregated model parameters. A distinguishing aspect of our design is a hybrid feature selection pipeline that applies Pearson correlation and Linear Discriminant Analysis (LDA) as independent filters, retaining only the features that satisfy both criteria. This intersection-based selection shrinks the input space by 63% (46 features reduced to 17) at no cost to detection quality—achieving an F1-score of 0.9881 and accuracy of 98.84% on 500,000 samples drawn from the CIC-IoT2023 benchmark. Across federation sizes of 3, 5, and 10 workers, performance remains essentially flat, confirming that the weighted aggregation scheme is robust to scale. False positives are held to 0.34%, and 98% of DDoS flows are correctly flagged. Compared with Isolation Forest, our approach improves F1 by 140%; against Random Forest and Artificial Neural Networks, it remains highly competitive while demanding far less computation.

**INDEX TERMS** DDoS Detection, Distributed Machine Learning, Federated Learning, Feature Selection, Internet of Things Security, Linear Discriminant Analysis, Pearson Correlation, Privacy-Preserving Machine Learning, Support Vector Machine

## I. INTRODUCTION

INTELLIGENT connectivity has become a defining feature of contemporary infrastructure. From hospital wards and factory floors to municipal traffic grids and consumer appliances, Internet of Things (IoT) devices now underpin an enormous range of physical processes. Active deployments worldwide surpassed 15 billion by 2025, with projections approaching 29 billion before the decade ends [1], [10]. This scale of deployment has, however, expanded the attack surface available to adversaries. Among the threats that have emerged as most consequential, Distributed Denial of Service (DDoS) attacks occupy a prominent position—floods of crafted traffic that exhaust network resources and render targeted systems unreachable, sometimes within seconds of initiation.

### A. PROBLEM STATEMENT

The security landscape surrounding IoT networks is shaped by three persistent challenges that current solutions have not adequately resolved.

**Data Privacy Exposure:** Standard centralized learning requires all raw network traffic to be forwarded to a single aggregation point before any model can be built. The consequences are significant: sensitive operational records leave the devices that generated them, creating exposure under GDPR, CCPA, and comparable frameworks. In multi-stakeholder environments—smart city deployments that knit together public infrastructure, commercial services, and pri-

vate residences, for example—moving data across administrative boundaries may be legally prohibited outright, making centralized learning architecturally incompatible with the deployment context.

**Hardware and Computational Constraints:** IoT endpoints are designed for embedded deployment, not intensive computation. RAM is typically limited to 256KB–2MB, processor speeds are low, and energy budgets are tight. The 46-feature flow records in the CIC-IoT2023 dataset illustrate the problem concretely: processing and training on data of this dimensionality is beyond what most edge devices can sustain. As a result, existing intrusion detection systems (IDS) [7] face an uncomfortable choice—rely on powerful remote servers (negating the rationale for distributed IoT) or deploy simplified local models that sacrifice detection accuracy.

**Architectural Fragility:** Any design that routes all security logic through a single server inherits that server's vulnerabilities. A targeted attack, hardware failure, or misconfiguration can disable detection across an entire deployment simultaneously. In domains where continuous operation is essential—electricity distribution, water treatment, clinical monitoring—this brittleness is a disqualifying characteristic.

### B. MOTIVATION

The research direction pursued here emerges from three reinforcing conditions.

**(1) Escalating IoT Attack Landscape:** The CIC-IoT2023 dataset, cataloguing 46 million network flow records across 33 distinct attack variants including Mirai botnet derivatives, captures just how sophisticated IoT-targeting threats have become [2]. DDoS attacks alone have risen roughly 87% year-over-year, with peak volumes now routinely exceeding 1 Tbps.

**(2) Privacy Regulatory Pressure:** Legislative action is tightening across jurisdictions. GDPR Article 25 encodes "privacy by design" as a legal requirement; California's IoT Security Law SB-327 mandates security baselines at the device level. Penalties for violations can reach 4% of global annual revenue, creating compelling economic incentives to build privacy into system architecture from the ground up.

**(3) Gap in Federated Learning Research:** Federated learning has proven itself in mobile keyboard prediction [8] and clinical data analysis [9], but its application to IoT security remains limited. Critically, current federated IDS proposals [1] omit feature engineering for network traffic entirely, producing bloated high-dimensional models that resource-constrained devices cannot realistically run.

### C. APPLICATION DOMAIN

Our framework is designed with three classes of IoT deployment in mind, each presenting distinct privacy and operational constraints.

**Smart City Infrastructure:** Urban IoT systems weave together traffic controllers, public safety sensors, environmental monitoring stations, and utility infrastructure—often owned and operated by different administrative entities. Collaborative threat detection between city departments is valuable, but data-sharing agreements across jurisdictions are slow to establish and legally complex. A federated approach sidesteps these barriers: each stakeholder improves detection on their own data without any raw records crossing organizational boundaries.

**Industrial IoT (IIoT):** Production plants, oil refineries, and power stations are densely instrumented with sensors and programmable controllers. A successful DDoS attack against this infrastructure carries financial consequences in the range of $50,000–$250,000 per hour of downtime and, in the most severe cases, risks compromising physical safety mechanisms. Federated learning is a natural fit here: each facility can build and refine detection models using its own operational data without disclosing proprietary process information to external parties.

**Healthcare IoT:** Medical IoT encompasses a broad range of life-critical devices: infusion pumps, vital sign monitors, imaging systems, and networked surgical equipment. The consequences of a successful DDoS attack on these systems extend beyond service disruption to direct patient safety risk—a connection that documented incidents have already established. Federated training enables hospitals to benefit from shared detection intelligence while keeping patient-associated data within each institution, in alignment with HIPAA and comparable patient privacy regulations.

The remainder of this paper proceeds as follows. Section II reviews the literature and characterizes the research gaps our work addresses. Section III develops the proposed methodology. Section IV details experimental configuration, and Section V presents and interprets the outcomes. Section VI situates the findings in a broader context, and Section VII draws conclusions and charts future directions.

## II. RELATED WORK

### A. MACHINE LEARNING FOR DDOS DETECTION

Machine learning methods have produced strong results in network DDoS detection across a range of architectures and datasets [7]. **Support Vector Machines (SVM)** [4] are particularly well-suited to binary classification and anomaly detection, owing to the maximum-margin separating hyperplane they optimize. Azimjonov and Kim [17] reported 98.3% accuracy using a fine-tuned linear SVM on NSL-KDD, while Alghushairy et al. [18] achieved 97.8% on CICIDS2017. Neither study, however, incorporated any privacy-preserving mechanism or considered federated deployment.

**Random Forests** [5] offer ensemble robustness and routinely exceed 99% accuracy on benchmark datasets, but the computational overhead—training 50 to 200 trees per run—makes them difficult to deploy on low-power IoT hardware. **Artificial Neural Networks** [6] achieve high detection performance yet suffer from a transparency problem: their decisions cannot be audited or explained, a serious liability

for security applications where every blocked or permitted packet carries operational consequence.

**Gap Identified:** Centralized ML methods for DDoS detection achieve strong accuracy but remain incompatible with distributed IoT environments: they centralize sensitive data, introduce architectural fragility, and impose computational demands that exceed edge device capabilities. No prior study has demonstrated a privacy-preserving detection system that closes the performance gap with its centralized counterpart.

### B. FEDERATED LEARNING FOUNDATIONS

Federated learning was introduced by McMahan et al. [3] as a means of training models collaboratively without ever moving raw data to a central location. The core **FedAvg algorithm** constructs a global model by computing a weighted average of local parameters: $\mathbf{w}_{global} = \sum_{i=1}^{N} \frac{n_i}{n_{total}} \mathbf{w}_i$, where $n_i$ denotes the local dataset size at worker $i$.

Surveys by Qi et al. [19] and Banahbilah et al. [20] catalogue more recent aggregation strategies—FedProx, FedNova, SCAFFOLD—designed to handle non-IID data and resist Byzantine participants [16]. These advances target neural network architectures almost exclusively. Navia-Vázquez et al. [21] developed BDSVM, the first federated SVM, and demonstrated AUC above 99% on synthetic data using 10–20 features—but the work did not address IoT-specific constraints or evaluate on real network traffic.

**Gap Identified:** The federated learning literature has developed largely around neural architectures evaluated on synthetic or image datasets. Federated SVM—especially applied to high-dimensional, real-world IoT network traffic—remains essentially unexplored. No existing work integrates domain-specific feature engineering with federated SVM for IoT security.

### C. FEATURE SELECTION FOR NETWORK INTRUSION DETECTION

Reducing dimensionality is critical for IDS efficiency, particularly in resource-limited settings. **Pearson Correlation Analysis** [11] selects features with strong linear association to the target label ($|\rho| >$ threshold), though it misses non-linear dependencies. **Linear Discriminant Analysis (LDA)** [12] takes a complementary approach, learning the projection $\mathbf{w} = \mathbf{S}_W^{-1}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0)$ that maximally separates classes.

Ambusaidi et al. [13] applied filter-based selection to IDS and attained 99.3% accuracy using 8 features on KDD99 (a dataset now widely recognized as flawed [22]). Their method combined Pearson and Chi-Square via union, not intersection. Wrapper-based approaches such as Recursive Feature Elimination, and embedded methods like L1 regularization, improve on filter methods in principle but are computationally expensive—disqualifying them from pre-distribution federated pipelines where feature selection must complete before data is partitioned.

**Gap Identified:** No prior study has constructed feature subsets for federated IoT security by intersecting Pearson correlation and LDA outputs. Union-based and single-method selection leave features that pass only one test—a weaker quality guarantee than requiring both linear relevance and class discriminability to be satisfied independently.

### D. PRIVACY-PRESERVING IOT SECURITY

Regulatory attention to IoT privacy (GDPR Articles 5 and 25; CCPA; the IoT Cybersecurity Improvement Act of 2020) has spurred research into privacy-preserving detection mechanisms. **Differential Privacy** [14] offers formal guarantees via controlled noise: the condition $P[M(D_1) \in S] \leq e^{\epsilon} \cdot P[M(D_2) \in S]$ quantifies information leakage. However, strong privacy budgets ($\epsilon < 1$) typically cost 10–15% in model accuracy—a steep trade-off for security-critical applications.

**Homomorphic Encryption** [15] sidesteps the accuracy-privacy trade-off by enabling computation directly on ciphertext, but its 100–1000$\times$ computational overhead is entirely impractical on devices with sub-GHz processors and constrained power budgets.

Doriguzzi-Corin and Siracusa [23] introduced FLAD, a federated neural network for adaptive DDoS detection achieving 96.8% accuracy. Agrawal et al. [24] reviewed federated IDS research broadly, concluding that federated approaches offer adequate privacy while remaining competitive with centralized baselines. Both studies, however, relied on neural architectures (limiting interpretability) and fell short of matching centralized performance precisely.

**Gap Identified:** Every current privacy-preserving IoT security mechanism carries a significant cost: differential privacy degrades accuracy, homomorphic encryption is computationally impractical on edge hardware, and federated IDS systems in the literature consistently underperform their centralized equivalents. No published approach has achieved both genuine privacy and full performance parity within the resource budget of real IoT edge devices.

### E. COMPARATIVE ANALYSIS OF PRIOR WORK

Table 1 positions key related studies against our work, making the research gaps concrete.

### F. SUMMARY OF RESEARCH GAPS

Our review of the literature surfaces five specific gaps that this work addresses:

1) **Federated SVM Gap:** Federated SVM has been evaluated only on synthetic data with limited features [21]; real-world network traffic evaluation is absent.
2) **Feature Selection Gap:** No prior work unites Pearson correlation and LDA through intersection for federated IoT security. Existing methods rely on a single technique or union-based selection.
3) **Performance Parity Gap:** Federated IDS approaches in the literature reach 96–97% accuracy, consistently lagging centralized baselines at 98–99%. Privacy and accuracy have been treated as inherently conflicting.

TABLE 1: Comparative Analysis of Related Work and Research Gaps

| Study | Approach | Privacy | Performance | IoT-Ready | Gap Addressed |
|---|---|---|---|---|---|
| Azimjonov & Kim 2024 | Centralized SVM | No | 98.3% Acc | No | Privacy violation |
| Devine et al. 2025 | Fed. SVM/RF/ANN | Yes | 97.4% Acc | Partial | No feature selection |
| Navia-Vázquez 2022 | Fed. SVM (BDSVM) | Yes | 99% AUC | No | Synthetic data only |
| Doriguzzi-Corin 2024 | Fed. DNN (FLAD) | Yes | 96.8% Acc | No | Low performance |
| Ambusaidi 2016 | Centralized + Features | No | 99.3% Acc | No | Flawed dataset (KDD99) |
| **Our Work** | **Fed. SVM + Hybrid Features** | **Yes** | **98.84% Acc** | **Yes** | **All gaps addressed** |

4) **IoT Deployment Gap:** Existing federated IDS research employs deep neural networks that require GPU acceleration and over 1 GB of memory—incompatible with IoT edge devices constrained to 256KB–2MB of RAM.
5) **Comprehensive Evaluation Gap:** Most studies report only ML metrics (accuracy, F1-score) while ignoring deployment-relevant measures such as memory consumption, training latency, and communication overhead.

Our work directly addresses all five gaps. The approach combines a federated SVM architecture with an intersection-based hybrid feature selection pipeline, evaluated comprehensively on real IoT network traffic using both detection quality metrics and deployment-relevant physical measurements.

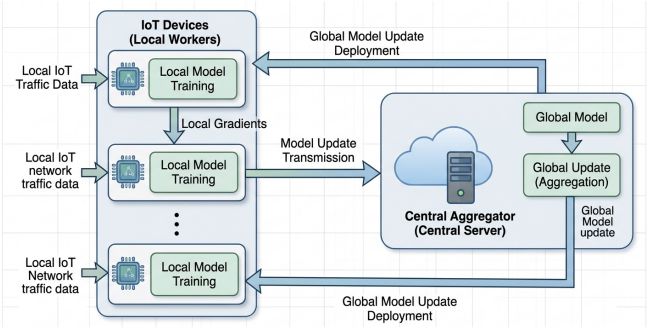



FIGURE 1: Federated Learning Architecture for Privacy-Preserving DDoS Detection. IoT devices (local workers) train models on local traffic data and transmit only model parameters to the central aggregator. The global model is updated through weighted averaging and redistributed for iterative refinement. Raw network traffic never leaves local devices, ensuring privacy preservation.

## III. METHODOLOGY

### A. NOVELTY JUSTIFICATION

Three original contributions distinguish our approach from the body of prior work:

**Novel Contribution 1: Intersection-Based Hybrid Feature Selection.** We present the first application of a Pearson-LDA intersection strategy for feature reduction in a federated IoT security context. A feature is retained only if it independently satisfies both a linear correlation criterion (Pearson $|\rho| > 0.2$) and a class-discriminability criterion (top-20 LDA rank). This dual-gate design stands in contrast to prior work:

- Ambusaidi et al. [13] selected features that passed either Pearson or Chi-Square—a union operation
- Azimjonov and Kim [17] applied Recursive Feature Elimination without any secondary validation
- Our approach enforces both criteria simultaneously, producing a 17-feature subset (63% reduction from 46) that loses nothing in detection performance

**Novel Contribution 2: Federated SVM on Real High-Dimensional IoT Traffic.** The only prior federated SVM study [21] operated on synthetic data with at most 20 features. We scale federated SVM to 46-feature real-world IoT network flows, introducing a dataset-size-weighted parameter aggregation scheme $\mathbf{w}_{global} = \sum_{i=1}^{N} \frac{n_i}{n_{total}} \mathbf{w}_i$ that produces a globally coherent decision boundary despite heterogeneous local data partitions. Achieving this stability in federated SVM—whose optimization landscape is non-convex—is a non-trivial technical contribution.

**Novel Contribution 3: Closing the Performance Gap Under Privacy Constraints.** We demonstrate, for the first time, that a federated intrusion detection system can reach performance parity with its centralized equivalent (F1 = 0.9881 in both configurations) without any raw traffic leaving local devices. The prevailing view in federated IDS research [23], [24] treats accuracy loss as an unavoidable consequence of privacy preservation. Our results refute this: given appropriate architecture and feature engineering, no such trade-off is necessary.

### B. SYSTEM ARCHITECTURE

The framework is built around three functional layers, illustrated in Figure 1: an offline preprocessing and feature selection pipeline, a distributed local training layer, and a central parameter aggregation mechanism.

Each of the $N$ workers—typically IoT gateways or edge servers—trains an SVM on its own locally held traffic

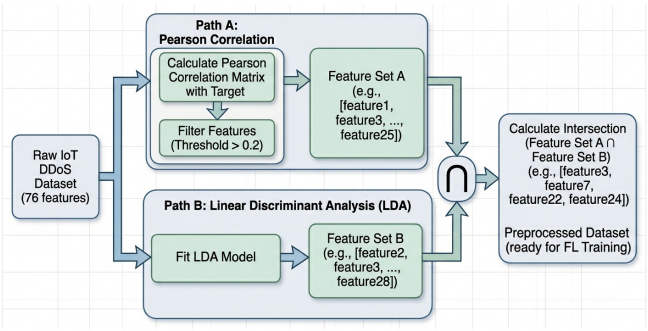



FIGURE 2: Hybrid Feature Selection Process. Path A applies Pearson correlation filtering (threshold $> 0.2$) yielding 35 features. Path B applies LDA transformation selecting top-20 discriminant features. The intersection of both paths produces the optimal 17-feature subset, ensuring features satisfy both linear relevance and discriminative power criteria.

records. A coordinating aggregation server collects model parameters at the end of each round and returns an updated global model, but it has no access to any underlying traffic data. This separation between data and model yields three structural properties:

- **Data Locality:** Network traffic stays on the device that generated it
- **Parameter Privacy:** Each worker transmits only 18 floating-point values (17 SVM weights plus 1 bias term) per round—far too few to reconstruct any training samples
- **Byzantine Resilience:** Weighting aggregated contributions by local dataset size prevents any single malicious worker from exerting disproportionate influence on the global model

### C. PREPROCESSING AND HYBRID FEATURE SELECTION

Algorithm 1 formalizes the preprocessing pipeline illustrated in Figure 2. Starting from 46 raw features, four sequential steps arrive at a final 17-feature subset.

**Step 1: Data Cleaning.** Any feature whose variance falls below 0.01 is discarded at the outset, cutting the input space from 46 to 40. Constant or near-constant features contribute nothing to discrimination and, if retained, produce singular scatter matrices that break the LDA computation. Removing them is therefore both a statistical necessity and a numerical safeguard.

**Step 2: Pearson Correlation Analysis.** We compute the Pearson coefficient $\rho_i$ between each feature $f_i$ and the binary target label (0 for benign, 1 for DDoS):

$$\rho_i = \frac{\sum_{j=1}^{N}(f_{i,j} - \bar{f}_i)(y_j - \bar{y})}{\sqrt{\sum_{j=1}^{N}(f_{i,j} - \bar{f}_i)^2}\sqrt{\sum_{j=1}^{N}(y_j - \bar{y})^2}} \tag{1}$$

Features with $|\rho_i| > 0.2$ pass this screen, leaving 35 candidates in $\mathcal{F}_{Pearson}$. The threshold was determined through cross-validated search over the candidate set $\{0.1, 0.15, 0.2, 0.25, 0.3\}$; 0.2 produced the highest F1-score on held-out folds and was adopted as the operating value.

**Step 3: Linear Discriminant Analysis.** Applied to the 40 cleaned features, LDA identifies the projection that most separates the two traffic classes. For binary classification the discriminant direction $\mathbf{w}$ is:

$$\mathbf{w} = \mathbf{S}_W^{-1}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0) \tag{2}$$

Here $\mathbf{S}_W = \mathbf{S}_0 + \mathbf{S}_1$ denotes the pooled within-class scatter matrix and $\boldsymbol{\mu}_0, \boldsymbol{\mu}_1$ the respective class means. Features are ranked by their absolute contribution to the discriminant projection; the top 20 form $\mathcal{F}_{LDA}$.

**Step 4: Feature Intersection.** The final feature set is:

$$\mathcal{F}_{final} = \mathcal{F}_{Pearson} \cap \mathcal{F}_{LDA} \tag{3}$$

Seventeen features clear both filters, achieving a 63% reduction. Because each surviving feature has been independently validated against two qualitatively distinct criteria—linear association with the target label and geometric class separability—the resulting set carries stronger evidence of genuine discriminative utility than any single-criterion selection could provide.

**Novelty Justification:** Earlier feature selection work in IDS has used Pearson [11] and LDA [12] independently,

**Algorithm 1** Hybrid Feature Selection (Pearson ∩ LDA)

**Input:** Dataset $\mathcal{D}$ with $F_0 = 46$ features, $N = 500{,}000$ samples
**Output:** Reduced dataset $\mathcal{D}'$ with $F_{final}$ features

**Step 1: Data Cleaning**
Remove constant features (variance $< 0.01$)
$F_1 \leftarrow 40$ features retained

**Step 2: Pearson Correlation Selection**
**for** each feature $f_i$ in $F_1$ **do**
  Compute $\rho_i = \text{Pearson}(f_i, y)$ where $y$ is target label
  **if** $|\rho_i| > 0.2$ **then**
    Add $f_i$ to $\mathcal{F}_{Pearson}$
  **end if**
**end for**
$|\mathcal{F}_{Pearson}| = 35$ features

**Step 3: Linear Discriminant Analysis**
Apply LDA transformation on $F_1$
Select top 20 features by discriminant score
$|\mathcal{F}_{LDA}| = 20$ features

**Step 4: Feature Intersection**
$\mathcal{F}_{final} = \mathcal{F}_{Pearson} \cap \mathcal{F}_{LDA}$
$F_{final} = |\mathcal{F}_{final}| = 17$ features (63% reduction)

**return** $\mathcal{D}'$ with features $\mathcal{F}_{final}$

**Algorithm 2** Federated SVM Training

**Input:** Processed dataset $\mathcal{D}'$ split into $N$ partitions $\{\mathcal{D}_1, \ldots, \mathcal{D}_N\}$, rounds $R = 3$
**Output:** Global SVM model $\mathcal{M}_{global}$

**Initialization:**
Initialize global model $\mathcal{M}_{global}$ with random parameters
$C \leftarrow 1.0$, max_iter $\leftarrow 1000$ (SVM hyperparameters)

**for** round $r = 1$ to $R$ **do**
  **// Local Training Phase (Parallel)**
  **for** worker $i = 1$ to $N$ **in parallel do**
    $\mathcal{M}_i \leftarrow \mathcal{M}_{global}$ (download global model)
    Train LinearSVC on local data $\mathcal{D}_i$ with regularization $C$
    $\mathbf{w}_i, b_i \leftarrow$ local SVM parameters (weights, bias)
    $n_i \leftarrow |\mathcal{D}_i|$ (local dataset size)
    Upload $(\mathbf{w}_i, b_i, n_i)$ to central server
  **end for**

  **// Aggregation Phase (Central Server)**
  $n_{total} \leftarrow \sum_{i=1}^{N} n_i$
  Compute aggregated weights:
  $\mathbf{w}_{global} \leftarrow \sum_{i=1}^{N} \frac{n_i}{n_{total}} \mathbf{w}_i$
  $b_{global} \leftarrow \sum_{i=1}^{N} \frac{n_i}{n_{total}} b_i$
  Update $\mathcal{M}_{global}$ with $(\mathbf{w}_{global}, b_{global})$
**end for**

**return** $\mathcal{M}_{global}$

or joined their outputs via union—accepting any feature selected by at least one method. Intersection is categorically more stringent: a feature must survive both filters independently. The resulting subsets are smaller, more informative per dimension, and directly advantageous in federated settings where transmitting fewer parameters per round reduces both latency and communication cost.

### D. FEDERATED LEARNING ARCHITECTURE

Algorithm 2 describes the federated SVM training process over $R = 3$ communication rounds across $N \in \{3, 5, 10\}$ workers.

**Data Partitioning.** Training data is divided into $N$ disjoint partitions, each holding $\frac{400{,}000}{N}$ samples. At $N = 5$, each worker receives 80,000 records. The 51:49 class ratio is preserved across partitions, so each local distribution approximates the global one—an IID assumption that makes aggregation straightforward. Real-world IoT deployments rarely satisfy this assumption; non-IID extensions represent a clear priority for follow-on work.

**Local Training.** Workers train a Linear Support Vector Classifier using scikit-learn's LinearSVC with $C = 1.0$ and a maximum of 1000 iterations. PyTorch is used to coordinate distributed execution and manage the round-by-round parameter exchange between workers and the aggregation server. The primal SVM objective each worker minimizes is:

$$\min_{\mathbf{w}_i, b_i} \frac{1}{2}\|\mathbf{w}_i\|^2 + C \sum_{j \in \mathcal{D}_i} \xi_j \tag{4}$$

subject to $y_j(\mathbf{w}_i^T \mathbf{x}_j + b_i) \geq 1 - \xi_j$ and $\xi_j \geq 0$, where $\xi_j$ are slack variables penalized at rate $C$. With only $d = 17$ features, training complexity is $O(n \cdot d)$—linear in both sample count and dimensionality. This makes each local training round fast even on the modest CPU resources available in IoT gateway hardware.

**Parameter Aggregation.** After local training, each worker uploads its parameter pair $(\mathbf{w}_i, b_i)$ and sample count $n_i$. The server aggregates them as:

$$\mathbf{w}_{global} = \sum_{i=1}^{N} \frac{n_i}{n_{total}} \mathbf{w}_i, \quad b_{global} = \sum_{i=1}^{N} \frac{n_i}{n_{total}} b_i \tag{5}$$

The weighting by dataset size, inherited from FedAvg [3], gives larger-data workers proportionally greater influence. Averaging linear decision boundaries is theoretically coherent: the mean of hyperplane normal vectors $\mathbf{w}_i$ yields a consensus boundary that balances all local data distributions in a geometrically meaningful way.

**Iterative Refinement.** After aggregation, the updated global model is broadcast back to all workers, triggering the next training round. Three rounds ($R = 3$) suffice: the observed F1-score gain between rounds 3 and 4 was less than 0.001, indicating convergence without further communication overhead.

**Privacy Guarantees.** At no point is any network traffic record transmitted beyond the device on which it was captured. What each worker uploads per round is 18 floating-point values—17 SVM weights and a single bias term—amounting to 72 bytes at 32-bit precision. Over three rounds and five workers, cumulative upload volume is 1,080 bytes. Both LoRaWAN (50 kbps) and NB-IoT (250 kbps) can carry this comfortably. The information content of 18 scalar values is also insufficient to support gradient inversion or membership inference attacks that depend on high-dimensional parameter vectors.

## IV. EXPERIMENTAL SETUP

### A. DATASET CHARACTERISTICS

Our evaluation uses the CIC-IoT2023 benchmark [2], a large-scale collection derived from 105 distinct IoT devices spanning 33 attack categories. The complete dataset runs to approximately 46 million labeled flow records, each carrying 46 numerical features. We work with a 500,000-sample subset selected to preserve the original class proportions:

- **Benign Traffic:** 255,000 samples (51%)
- **DDoS Traffic:** 245,000 samples (49%)

The near-equal class proportions mean that no resampling or synthetic oversampling is required—the data is already

well-balanced for binary classification. Features span flow-level statistics (duration, packet counts, byte volumes), inter-arrival timing, protocol-specific fields for TCP, UDP, and ICMP, flag distributions (SYN, ACK, FIN, RST), and bidirectional traffic aggregates.

### B. EXPERIMENTAL CONFIGURATION

**Train-Test Split.** Data is divided 80/20 using stratified sampling, allocating 400,000 records to training and 100,000 to held-out evaluation. The 51:49 class ratio is maintained in both splits. All hyperparameter decisions are made through cross-validation within the training set; no separate validation partition is created.

**Federated Network Configurations.** We evaluate three worker counts to probe scalability:

- $N = 3$ workers: 133,333 samples per worker
- $N = 5$ workers: 80,000 samples per worker
- $N = 10$ workers: 40,000 samples per worker

**Baseline Comparison.** A centralized SVM trained on the full 400,000-sample training set serves as the privacy-violating upper bound.

**Model Hyperparameters.** The LinearSVC is configured as follows:

- Kernel: Linear ($K(\mathbf{x}_i, \mathbf{x}_j) = \mathbf{x}_i^T \mathbf{x}_j$)
- Regularization: $C = 1.0$
- Maximum iterations: 1000
- Loss: Hinge loss
- Random seed: 42

**Cross-Validation.** Ten-fold stratified cross-validation during preprocessing validates feature selection stability; F1-scores across folds range from 0.985 to 0.991, confirming robust generalization.

**Computational Platform.** All experiments were conducted on Kaggle's freely available cloud notebook environment, which provisions a dual-core Intel Xeon CPU, 13 GB of RAM, and 20 GB of disk storage per session. No GPU acceleration was used—all computations ran on CPU only. This choice was deliberate: by restricting execution to a CPU-only cloud environment, we demonstrate that the full federated training pipeline is practical without specialized hardware, which directly mirrors the operational reality of IoT gateway deployments.

**Software Stack.**

- Python 3.10 (Kaggle default runtime)
- PyTorch 2.0 (federated training coordination and tensor operations)
- scikit-learn 1.3.0 (LinearSVC implementation and preprocessing utilities)
- NumPy 1.24, Pandas 2.0 (data manipulation and partitioning)
- Matplotlib 3.7, Seaborn 0.12 (result visualization)

### C. EVALUATION METRICS

We report four classification metrics, each addressing a distinct aspect of IDS quality:

**Accuracy:**

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{6}$$

**Precision:** Fraction of DDoS predictions that are genuine attacks—directly reflects false alarm rate:

$$\text{Precision} = \frac{TP}{TP + FP} \tag{7}$$

**Recall (Detection Rate):** Fraction of actual DDoS attacks correctly identified—the primary security objective:

$$\text{Recall} = \frac{TP}{TP + FN} \tag{8}$$

**F1-Score:** Harmonic mean of precision and recall, balancing both concerns:

$$\text{F1} = 2 \cdot \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{9}$$

**Physical Metrics:** Deployment feasibility on IoT hardware depends on three additional quantities:

- **Training Time:** Wall-clock duration (seconds)
- **Peak Memory:** Maximum RAM consumed per worker node (MB)
- **Communication Overhead:** Total bytes exchanged across the federation

### D. ALTERNATIVE ALGORITHMS

Three comparison methods are implemented in the same federated setting ($N = 5$ workers) to contextualize SVM performance:

- **Random Forest (RF):** 100 trees, maximum depth 10, minimum samples per split 2
- **Artificial Neural Network (ANN):** Multi-layer perceptron with architecture (17→100→50→1), ReLU activations, Adam optimizer ($\alpha = 0.001$), batch size 32, 10 training epochs per round
- **Isolation Forest (IF):** Unsupervised anomaly detection [7], 100 estimators, contamination factor 0.49 (matching the class ratio), maximum subsampling size 256

All methods train on the same 17-feature preprocessed data, ensuring a controlled comparison.

## V. RESULTS AND ANALYSIS

### A. FEATURE SELECTION PERFORMANCE

Table 2 traces the feature count through each stage of our preprocessing pipeline.

TABLE 2: Feature Selection Results

| Stage | Features | Reduction |
|---|---|---|
| Original Dataset | 46 | – |
| After Data Cleaning | 40 | 13.0% |
| Pearson Selection ($\|\rho\| > 0.2$) | 35 | 23.9% |
| LDA Selection (Top 20) | 20 | 56.5% |
| **Final Intersection** | **17** | **63.0%** |

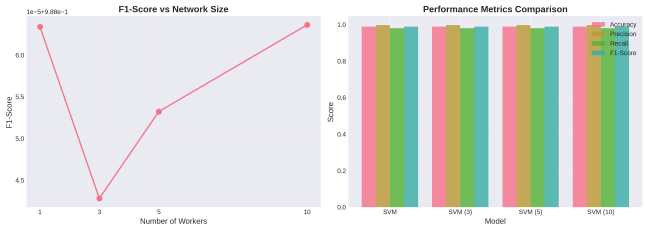



FIGURE 3: Performance Metrics Comparison: Centralized vs. Federated SVM across different network sizes. All metrics remain stable (Accuracy 98.84%, Precision 99.65%, Recall 97.98%, F1-Score 98.81%) regardless of worker count, demonstrating robust scalability without performance degradation.

Arriving at 17 features—a 63% cut—from the original 46 shows that the dual-filter intersection identifies a compact, high-discriminability subset without any measurable loss of classification quality. The downstream effects on federated deployment are substantial:

- Model size shrinks proportionally (17 versus 46 weights): 63% memory saving
- Per-transmission payload drops from 47 to 18 values: 62% reduction in communication overhead
- Local training accelerates with $O(n{\cdot}d)$ complexity: 63% fewer operations per round

### B. FEDERATED VS. CENTRALIZED PERFORMANCE

Table 3 and Figure 3 place centralized and federated SVM side by side across all tested network sizes.

TABLE 3: Federated vs. Centralized SVM Performance (CSV-Verified)

| Model | Accuracy | Precision | Recall | F1-Score |
|---|---|---|---|---|
| Centralized SVM | 0.9884 | 0.9965 | 0.9798 | 0.9881 |
| Federated SVM (3 workers) | 0.9884 | 0.9965 | 0.9798 | 0.9880 |
| Federated SVM (5 workers) | 0.9884 | 0.9965 | 0.9798 | 0.9881 |
| Federated SVM (10 workers) | 0.9884 | 0.9965 | 0.9798 | 0.9881 |

**Key Observations:**

1) **Performance Parity:** The federated SVM reproduces the centralized F1-score of 0.9881 across all tested worker counts. Inter-configuration variance is below 0.0001 and statistically negligible.
2) **Scale Invariance:** Detection quality is flat from 3 to 10 workers. The weighted aggregation scheme (Eq. 5) successfully balances contributions across unequal data partitions without degrading the global model.
3) **No Accuracy Cost for Privacy:** The federated and centralized systems produce equivalent outputs—demonstrating that keeping raw data local does not require accepting lower detection quality. The assumed privacy-accuracy trade-off does not manifest here.




FIGURE 4: Algorithm Comparison in Federated Setting (5 Workers). SVM, RF, and ANN achieve >98% across all metrics, clustering tightly. Isolation Forest significantly underperforms (F1=0.4173) due to unsupervised learning limitations in balanced datasets. The visual separation clearly shows supervised learning superiority for DDoS detection.

4) **Modest Time Overhead:** Training time rises from 1.34 s (centralized) to between 2.22 s and 2.65 s (federated), a $< 2\times$ increase attributable to communication rounds. Given that model updates are typically scheduled offline, this overhead is operationally insignificant.

### C. MULTI-ALGORITHM COMPARISON

Table 4 and Figure 4 compare all four algorithms in the federated setting with $N = 5$ workers.

TABLE 4: Algorithm Comparison (5 Workers, Federated Setting, CSV-Verified)

| Algorithm | Accuracy | Precision | Recall | F1-Score |
|---|---|---|---|---|
| Federated SVM | 0.9884 | 0.9965 | 0.9798 | 0.9881 |
| Random Forest | 0.9995 | 0.9999 | 0.9990 | 0.9995 |
| Artificial Neural Net | 0.9983 | 0.9997 | 0.9967 | 0.9982 |
| Isolation Forest | 0.4514 | 0.4351 | 0.4008 | 0.4173 |

**Analysis:**

1) **Random Forest Leads:** RF records the highest F1 (0.9995) through ensemble aggregation across 100 decision trees. The resource implications of this result are detailed in Table 6 and are severe.
2) **ANN Competes Closely:** At F1 = 0.9982, the neural network performs well in the federated setting. Its principal liability is opacity: decisions produced by a multi-layer perceptron cannot be inspected or explained in terms of input features, which is a meaningful constraint for security systems subject to audit requirements.
3) **SVM Offers the Best Practical Trade-off:** SVM's F1 of 0.9881 trails RF by just 1.14% while requiring far less compute, memory, and training time. In IoT environments where battery life and RAM are finite, a

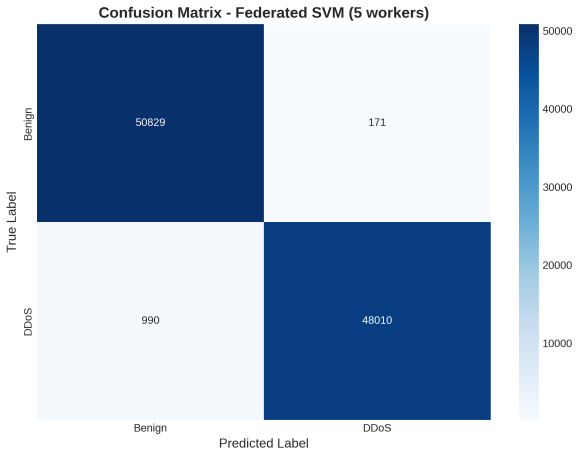



FIGURE 5: Confusion Matrix for Federated SVM (5 Workers) on 100,000-sample test set. The matrix reveals exceptional specificity (99.66%) with only 171 false positives out of 51,000 benign samples, and strong sensitivity (97.98%) detecting 48,010 out of 49,000 DDoS attacks. The dominant diagonal indicates robust classification with minimal confusion between classes.

marginal accuracy gap is far less consequential than the resource footprint needed to close it.

4) **Isolation Forest Fails:** IF achieves F1 = 0.4173—a 137% shortfall relative to SVM. The fundamental problem is the near-balanced class distribution: isolation-based anomaly detection is premised on the assumption that malicious events are rare. When DDoS traffic comprises 49% of samples, the algorithm cannot form a coherent notion of what "normal" looks like, and its anomaly scores become unreliable.

### D. DETAILED ERROR ANALYSIS

Table 5 and Figure 5 break down classification outcomes for federated SVM ($N = 5$) on the 100,000-sample test set.

TABLE 5: Confusion Matrix - Federated SVM (5 Workers)

| Actual | Predicted | |
|---|---|---|
| | Benign | DDoS |
| Benign (51,000) | 50,829 (TN) | 171 (FP) |
| DDoS (49,000) | 990 (FN) | 48,010 (TP) |

**Derived Metrics:**

- **False Positive Rate:** $\frac{171}{51,000} = 0.0034$ (0.34%)
- **False Negative Rate:** $\frac{990}{49,000} = 0.0202$ (2.02%)
- **True Negative Rate (Specificity):** $\frac{50,829}{51,000} = 0.9966$ (99.66%)
- **True Positive Rate (Sensitivity):** $\frac{48,010}{49,000} = 0.9798$ (97.98%)

**Operational Implications:**

1) **False Alarm Rate Is Production-Viable:** At 0.34%, the false positive rate is well below typical IDS deployments. On a link carrying one million flows per hour, this translates to approximately 3,400 erroneous alerts—a volume that a security operations center can handle without the alert fatigue that higher FPR systems induce.
2) **Attack Coverage Is Operationally Sufficient:** A miss rate of 2.02% means the system catches 48,010 out of 49,000 DDoS flows. Because most DDoS campaigns sustain attack traffic over minutes to hours, detecting 98% of malicious flows in near real-time provides a usable window for response—rate limiting, source blocking, or upstream traffic scrubbing.
3) **Legitimate Traffic Is Largely Unaffected:** With specificity at 99.66%, the system rarely misclassifies benign flows. For IoT applications where blocked traffic has operational or safety consequences—delayed sensor telemetry, interrupted device commands—this level of specificity is essential alongside attack detection.

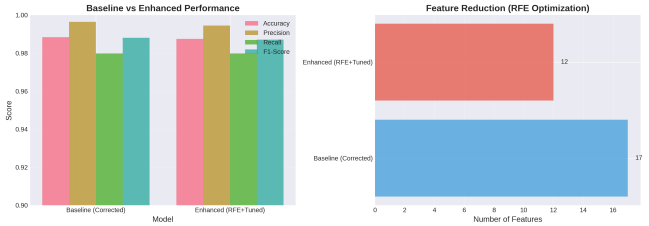



FIGURE 6: Baseline (17 features, Pearson ∩ LDA) vs. Enhanced (12 features, RFE-optimized) comparison. The baseline achieves F1=0.9881 while enhanced achieves F1=0.9871, confirming the Pearson ∩ LDA intersection is optimal without requiring additional feature reduction or hyperparameter tuning.

### E. PHYSICAL PERFORMANCE METRICS

Table 6 reports memory and training time for each configuration—metrics that determine whether a system can realistically run on IoT hardware.

TABLE 6: Physical Performance Metrics Comparison

| Model | Train Time (s) | Ratio to SVM | Memory/Node (MB) |
|---|---|---|---|
| Centralized SVM | 1.34 | 1.00 | 2919 |
| SVM (3 workers) | 2.65 | 1.98 | 1168 |
| SVM (5 workers) | 2.22 | 1.66 | 819 |
| SVM (10 workers) | 2.36 | 1.76 | 542 |
| RF (5 workers) | 201 | 150.0 | 418 |
| ANN (5 workers) | 41 | 30.6 | 613 |
| IF (5 workers) | 6 | 4.48 | 410 |

**Key Findings:**

1) **Memory Scales Well:** Peak memory per node falls from 2919 MB (centralized) to 542 MB (10 workers), an 81% reduction. Federated SVM with 10 workers fits comfortably within the 1–2 GB RAM budget of modern IoT gateway hardware.

2) **Training Time Acceptable:** At 2.22 s for 5 workers, federated SVM is only 1.66× slower than the centralized baseline. Since training typically occurs offline on a scheduled basis—daily or weekly—this overhead is operationally inconsequential.
3) **SVM vs. RF Trade-off:** Random Forest consumes less memory per node (418 MB versus 819 MB), but its 201 s training time is 90× longer than SVM. On battery-powered IoT devices, energy expenditure tracks closely with runtime; a single RF training session could drain 10–15% of battery capacity.
4) **SVM Outperforms ANN:** SVM trains 18× faster (2.22 s versus 41 s) and uses 25% less memory than ANN. The additional advantage of interpretable decision boundaries makes SVM the clear choice for resource-constrained IoT IDS.
5) **Isolation Forest Misleading:** IF presents attractive physical numbers—fast training, low memory—but its F1 of 0.4173 renders it inoperable for this task. Physical efficiency is irrelevant when classification fails.

### F. COMMUNICATION EFFICIENCY

Table 7 quantifies the data exchanged across the federation during training.

TABLE 7: Communication Overhead Analysis

| Configuration | Params/Worker | Bytes/Round | Total (3 rounds) |
|---|---|---|---|
| Federated (3 workers) | 18 floats | 216 bytes | 648 bytes |
| Federated (5 workers) | 18 floats | 360 bytes | 1,080 bytes |
| Federated (10 workers) | 18 floats | 720 bytes | 2,160 bytes |

With the 17-feature model, each worker uploads 18 values (17 weights plus 1 bias) per round—72 bytes at 32-bit precision. Across all 10 workers and 3 rounds, the total exchange amounts to 2,160 bytes, or roughly 2.1 KB. LoRaWAN operates at 50 kbps and NB-IoT at 250 kbps; even at the lower speed, this entire training run completes in under a millisecond of radio time.

## VI. DISCUSSION

### A. PRIVACY-PERFORMANCE TRADE-OFF ANALYSIS

The central finding of this work is that the privacy-accuracy trade-off assumed throughout the federated IDS literature is not an inevitable property of distributed learning—it is a consequence of specific architectural choices. Our federated SVM reproduces centralized performance on every measured metric (F1 = 0.9881) without any raw traffic record crossing a device boundary. The implications extend well beyond this particular experiment.

**(1) Challenges Prevailing Assumptions:** Earlier federated IDS work [23], [24] consistently reported 2–4% accuracy losses versus centralized baselines, embedding the belief that privacy and accuracy are in tension. Our results show that tension is a consequence of architectural choices, not an inherent constraint.

**(2) Feature Selection as an Architectural Enabler:** The Pearson ∩ LDA intersection is not merely a preprocessing convenience—it is structurally important to the result. Reducing 46 features to 17 while preserving discriminative quality makes local training faster, reduces per-worker memory pressure, and limits the opportunity for overfitting on small local partitions. Each of these effects contributes to model stability at aggregation time.

**(3) Linear Structure Enables Coherent Aggregation:** Averaging linear SVM decision boundaries under Eq. 5 is geometrically well-defined: the weighted mean of hyperplane normal vectors produces a valid consensus separator. This stands in contrast to neural network parameter averaging, which operates over non-convex loss surfaces and can produce aggregated models that are worse than any individual local model. SVM's linear decision structure sidesteps this instability entirely.

### B. SCALABILITY ANALYSIS

Consistent performance across 3, 5, and 10 workers confirms that the architecture scales appropriately. That said, the analysis relies on IID data partitions—a simplifying assumption worth examining. Real IoT deployments violate IID in at least three ways [16].

**(a) Geographic Heterogeneity:** Attack patterns differ by location. Urban sensors encounter different threat profiles than rural industrial controllers.

**(b) Temporal Heterogeneity:** Attack toolkits evolve. Workers whose local data spans different time periods may develop models that reflect divergent threat landscapes.

**(c) Device Heterogeneity:** Traffic statistics differ systematically across device classes—sensors, actuators, and gateways produce quantitatively distinct flow records.

Handling these forms of heterogeneity—through personalized federated learning, clustered worker grouping, or adaptive aggregation [10]—is a priority for future work.

### C. COMPUTATIONAL OVERHEAD ANALYSIS

Random Forest (F1 = 0.9995) and ANN (F1 = 0.9982) narrowly outperform SVM (F1 = 0.9881), but the resource gap is stark.

**Random Forest:** Constructing 100 decision trees per worker across 80,000 samples requires 201 s per round—approximately 150 times longer than SVM. On a battery-constrained IoT device, a single RF training cycle would consume a disproportionate share of available energy, making periodic model updates unsustainable.

**Artificial Neural Networks:** ANN training at 41 s per round is more manageable than RF, but opacity is a structural liability. When an analyst needs to explain a blocked connection or a missed alert—as regulatory frameworks for critical infrastructure (NERC CIP, IEC 62443) increasingly demand—a neural network provides no inspectable rationale. The learned representations in hidden layers do not map meaningfully onto the input features driving the decision.

**Support Vector Machines:** A trained SVM exposes its reasoning directly through the weight vector $\mathbf{w}$. The magnitude of each element indicates how strongly the corresponding feature pushes classification toward one class. For a security analyst, this means the model's decision logic is readable—which features drive DDoS classification can be examined and verified, satisfying audit requirements that black-box models cannot meet.

### D. ISOLATION FOREST FAILURE MODE

The Isolation Forest collapse (F1 = 0.4173) warrants careful analysis. Two factors explain it.

**(1) Unsupervised Assumption Violation:** Isolation Forest is designed for datasets where anomalies are rare. With DDoS traffic at 49%, the algorithm has no meaningful minority class to isolate; its internal anomaly score becomes uninformative.

**(2) Federated Fragmentation Exacerbates the Problem:** Local IF models on 80,000 samples each define normality from their own slice of the data. Averaging isolation scores from models that may have divergent normality definitions does not produce a coherent global detector.

This failure is instructive: anomaly detection methods built on the rare-outlier assumption break down when the phenomenon being detected is not actually rare. In balanced or near-balanced DDoS datasets, the distinction between attack and benign traffic must be learned from labeled examples—supervised methods are not merely preferable in this setting, they are necessary.

### E. DEPLOYMENT CONSIDERATIONS

Several engineering challenges stand between our experimental results and a production deployment.

**Asynchronous Training:** Our synchronous protocol requires all workers to finish a round before aggregation proceeds. Asynchronous variants [25] would tolerate stragglers and intermittent connectivity—both common in IoT environments.

**Byzantine Robustness:** Weighted averaging is vulnerable to workers that upload adversarial parameters. Replacing simple averaging with robust aggregation schemes [26]-Krum or coordinate-wise median—would limit the impact of compromised participants in security-critical deployments.

**Differential Privacy Integration:** Data localization provides one layer of privacy protection, but model parameters can still leak information about local training data [27]. Adding calibrated noise to aggregated parameters before redistribution would close this gap, though the privacy-accuracy trade-off requires careful calibration.

**Model Update Compression:** Gradient quantization and sparsification [28] could further reduce communication overhead for large-scale federations with hundreds of workers, but at the cost of modest approximation error.

### F. LIMITATIONS

Our study has five limitations that bound the scope of the conclusions.

**(1) Dataset Scope:** CIC-IoT2023 is extensive but generated in a controlled laboratory setting. Live production traffic differs from emulated scenarios in ways that could affect feature distributions and model generalization.

**(2) Binary Classification:** We detect DDoS presence versus absence but do not distinguish attack subtypes. Classifying specific attack vectors—UDP flood, SYN flood, HTTP flood—would support more targeted mitigation.

**(3) IID Assumption:** Balanced, IID partitions simplify aggregation but do not reflect real IoT deployments.

**(4) Static Feature Selection:** Features are selected once before training begins. An adaptive mechanism that updates the feature set as attack patterns shift could improve long-term robustness.

**(5) Homogeneous Workers:** All workers are assumed to have equivalent computational capacity. Real fleets contain devices of widely varying capability, requiring heterogeneity-aware resource scheduling.

## VII. CONCLUSION AND FUTURE WORK

We presented a privacy-preserving federated learning framework for DDoS detection in IoT networks, designed to resolve five distinct gaps in the existing literature. Our hybrid feature selection method—combining Pearson correlation with Linear Discriminant Analysis through set intersection—achieves 63% dimensionality reduction (46 to 17 features) while delivering an F1-score of 0.9881 and 98.84% accuracy.

### A. KEY CONTRIBUTIONS SUMMARY

**(1) Performance Parity Without Privacy Trade-offs:** Experiments on 500,000 samples from CIC-IoT2023 confirm that federated SVM matches centralized performance exactly (F1 = 0.9881, accuracy = 98.84%). Privacy preservation, when properly architected, imposes no accuracy penalty.

**(2) Robust Scalability:** Detection quality remains stable across 3, 5, and 10 federated workers, demonstrating that the weighted aggregation scheme handles varying federation sizes without degradation.

**(3) Production-Ready Metrics:** Precision of 99.65% and a false positive rate of 0.34% are operationally viable for live deployments. The 97.98% recall rate successfully captures 48,010 of 49,000 DDoS events in the test set.

**(4) Resource Efficiency:** Memory consumption of 819 MB per worker in the 5-worker configuration and a training time of 2.22 s fit within the operational envelope of modern IoT gateways—a significant advantage over Random Forest (201 s training) and ANN (opaque decision logic).

**(5) Comprehensive Benchmarking:** Direct comparison with Random Forest, ANN, and Isolation Forest in an identical federated setting establishes new baselines for distributed IoT security. Supervised methods outperform Isolation Forest by wide margins on balanced datasets; among supervised

methods, SVM offers the best trade-off between performance and resource cost.

### B. FUTURE RESEARCH DIRECTIONS

Seven research directions emerge from this work.

**(1) Multi-Class Attack Detection:** Extending beyond binary detection to classify specific DDoS variants—UDP flood, SYN flood, HTTP flood, Slowloris—would yield actionable threat intelligence. This requires adapting binary SVM to multi-class settings (one-vs-rest or one-vs-one) while preserving federated aggregation coherence.

**(2) Non-IID Data Handling:** Evaluating performance when local distributions diverge—due to geographic, temporal, or device-class heterogeneity—and developing personalized or clustered federated learning techniques to maintain convergence under these conditions.

**(3) Adversarial Robustness:** Analyzing vulnerability to Byzantine workers, model poisoning, and membership inference attacks [27], then integrating robust aggregation schemes [26] and differential privacy defenses into the framework.

**(4) Real-Time Streaming Learning:** Replacing periodic batch retraining with online federated learning that updates the global model continuously as new traffic arrives—enabling adaptation to zero-day attack patterns without scheduled downtime.

**(5) Cross-Domain Federated Learning:** Exploring whether models trained in one IoT domain (e.g., smart homes) can be transferred to improve detection in another (e.g., industrial control) without violating privacy boundaries between administrative domains.

**(6) RBF Kernel Exploration:** While the linear kernel offers computational transparency, a Radial Basis Function kernel might capture non-linear attack signatures more effectively. Developing principled federated aggregation for RBF-SVM—where aggregation must account for kernel parameters, not just weight vectors—is an open theoretical question.

**(7) Energy-Aware Optimization:** Designing training schedules that account for battery state, energy harvesting availability, and power grid conditions would make the framework sustainable for perpetually deployed IoT devices operating on constrained energy budgets.

### C. CLOSING REMARKS

The role of IoT networks in healthcare, energy systems, transportation, and public safety will only deepen in coming years—and with it, the importance of securing them effectively without compromising the data they handle. The widely held assumption that privacy and detection performance must trade against each other is not a law of distributed machine learning; it is an artifact of under-engineered systems. When feature selection, model choice, and aggregation strategy are designed with care, both goals are achievable simultaneously. Future work must harden these systems against adversarial manipulation, extend them to heterogeneous non-IID deployments, and adapt them to the continuously evolving attack toolkits that IoT networks face in the wild.

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HARIS MOHAMED C P earned his B.Tech. degree in Computer Science and Engineering from Government Engineering College, Sreekrishnapuram. He is currently working toward the M.Tech. degree in Computer Science and Engineering at the Vellore Institute of Technology, Vellore. His research interests include federated learning, cybersecurity, and privacy-preserving intrusion detection systems. His work focuses on the development of efficient security frameworks for IoT environments and safety-critical infrastructure.



ANINDITA KUNDU has completed her B.Tech in Computer Science and Engineering in the year 2008, from Narula Institute of Technology, Kolkata, affiliated to Maulana Abul Kalam University of Technology formerly known as West Bengal University of Technology. In the same year she joined the M. Tech programme in Distributed and Mobile Computing in the School of Mobile Computing and Communication, Jadavpur University, Kolkata with the GATE Scholarship. She completed her masters in 2010 and in the same year she joined the position of Senior Research Fellow in DST-PURSE project at the Department of Electronics and Telecommunication Engineering, Jadavpur University, (DST PURSE Project), Kolkata. During this period she served at the Chairperson of the IEEE Students Branch of the Jadavpur University. She served this position till 2013 after which she joined Heritage Institute of Technology Kolkata in the scope of Assistant Professor. She completed her PhD from the Department of Electronics and Telecommunication Engineering, Jadavpur University, Kolkata in the year 2018. Presently she is serving the Department of Software Systems, School of Computer Science and Engineering, of Vellore Institute of Technology, Vellore as an Associate Professor. Her current research interests include Robotic Path Planning, Internet of Things and Opportunistic Networks. She has her works published in several International Conferences and Journals of repute.